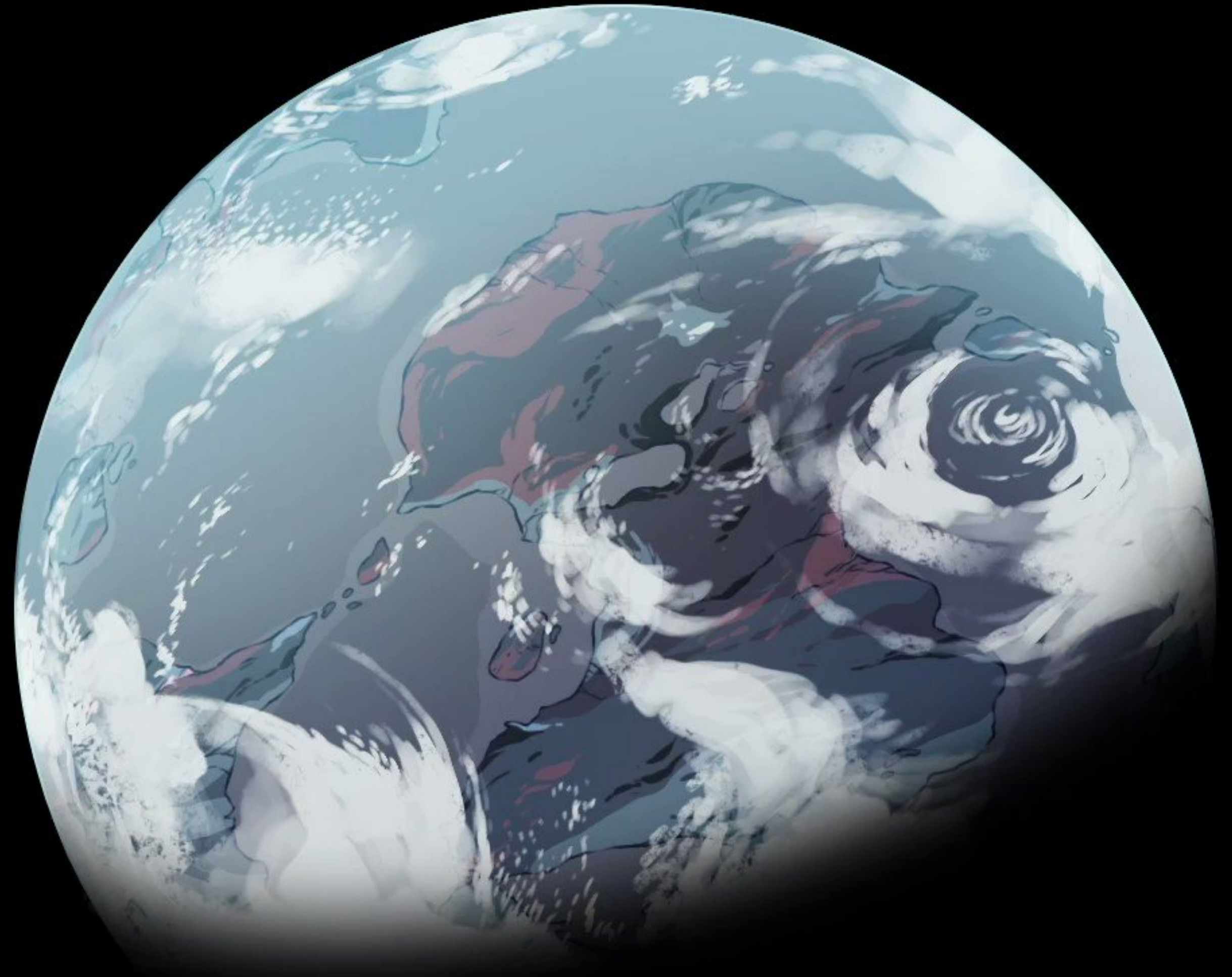


Science & Science Fiction

33-120 | Prof. Katie Breivik



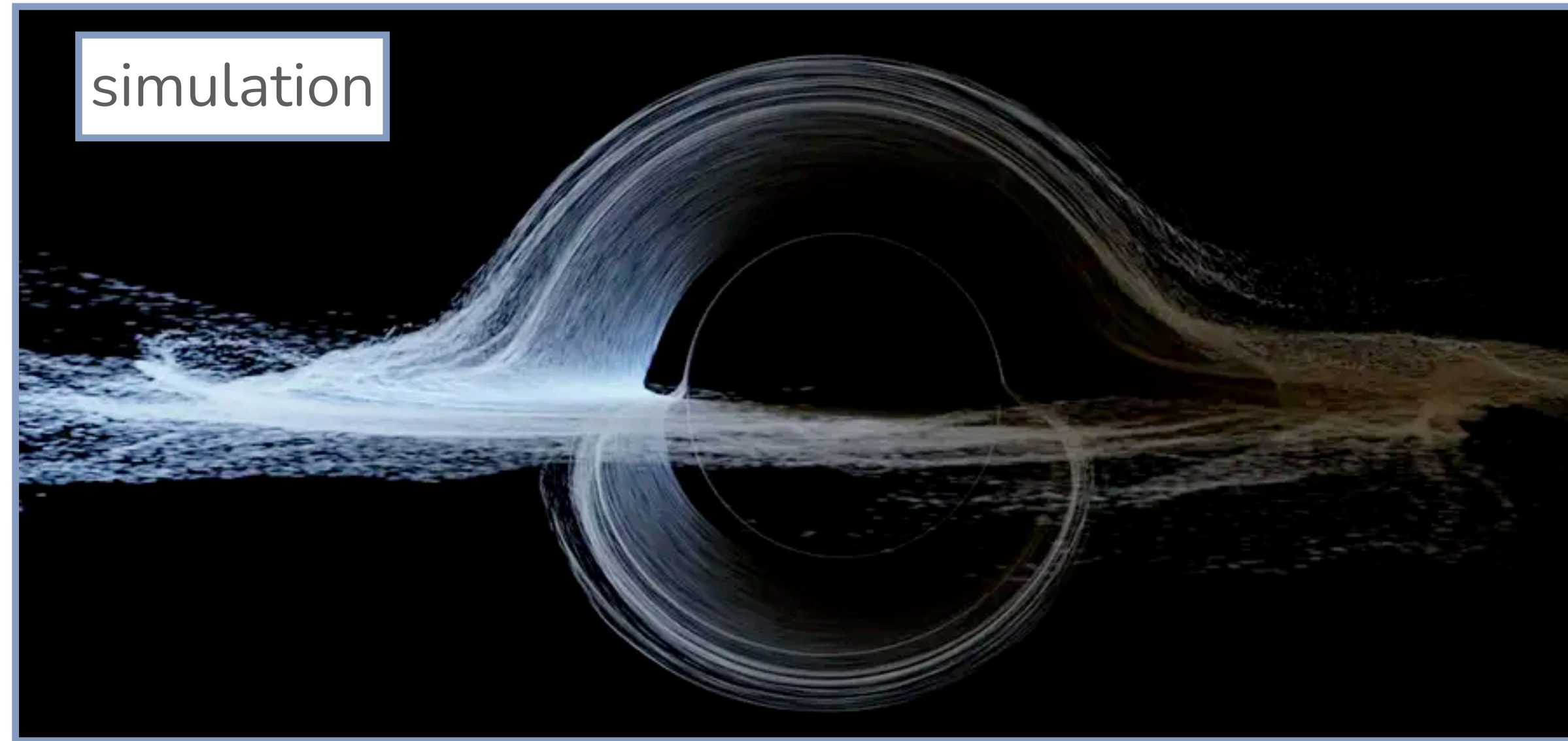
Announcements

- Problem 2 is due 01/31 at 11:59pm
- Exploration paper 1 is due 02/03 (tonight!)
 - Must be written in your own words!
- Team Project 1: due 02/05

Today:

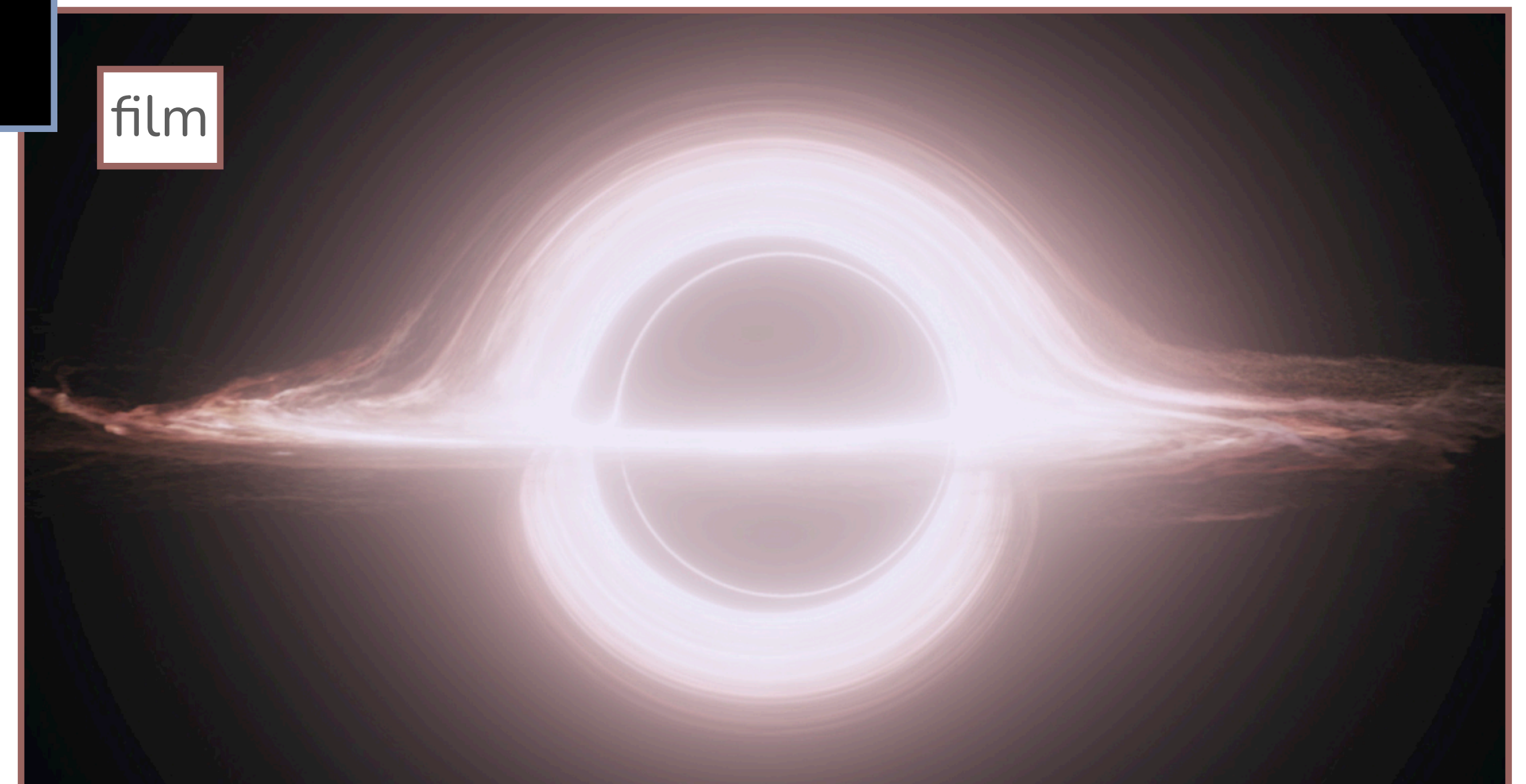
- More black holes!
- Gravitational waves

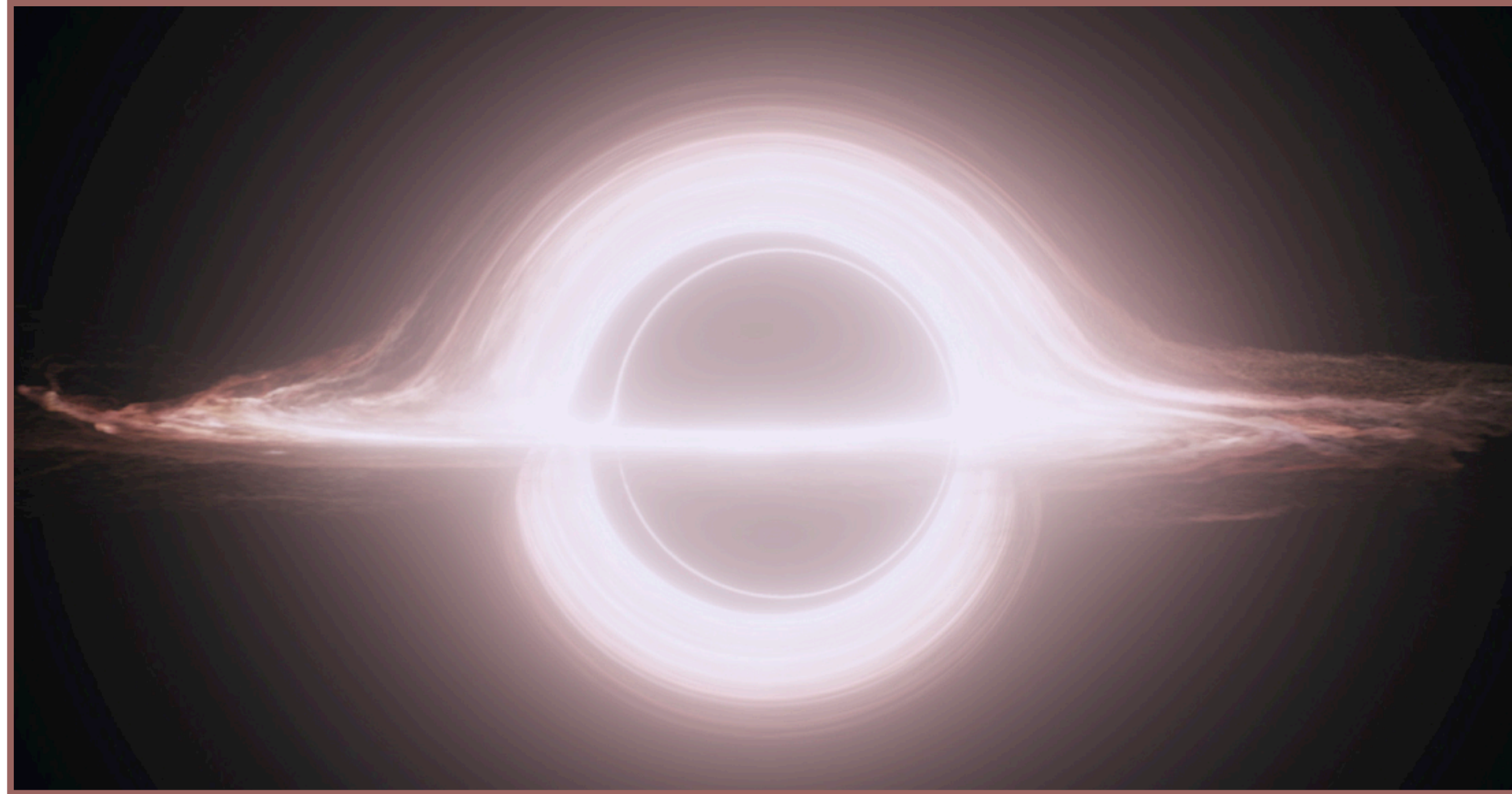
Astrophysical black holes in Science Fiction



“Gargantua” in Interstellar is the most faithful representation of a black hole in film to date

A collaboration between scientists, led by Kip Thorne, & artists!



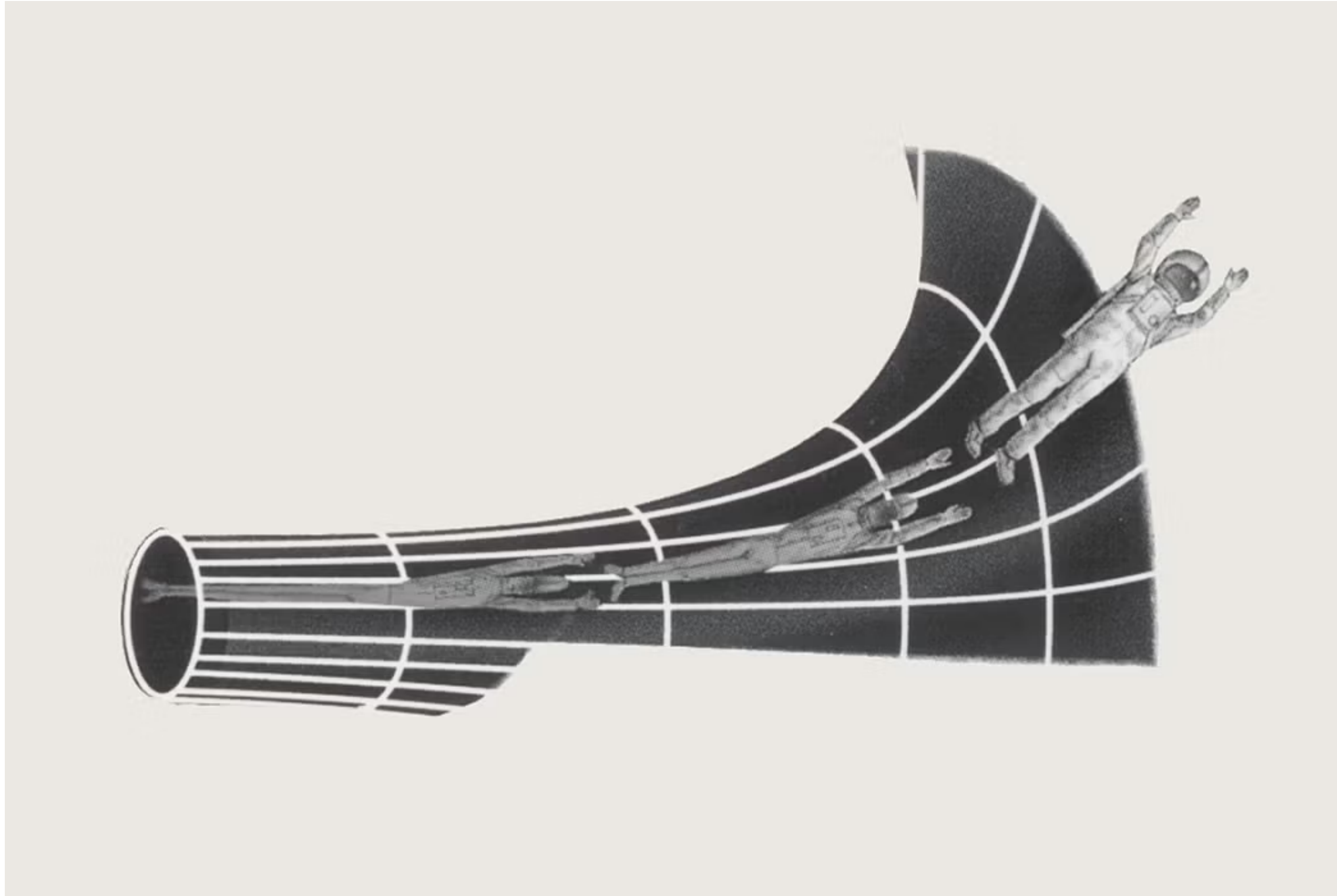


Interstellar

Directed:
Christopher Nolan

Warner Brothers
(2009)

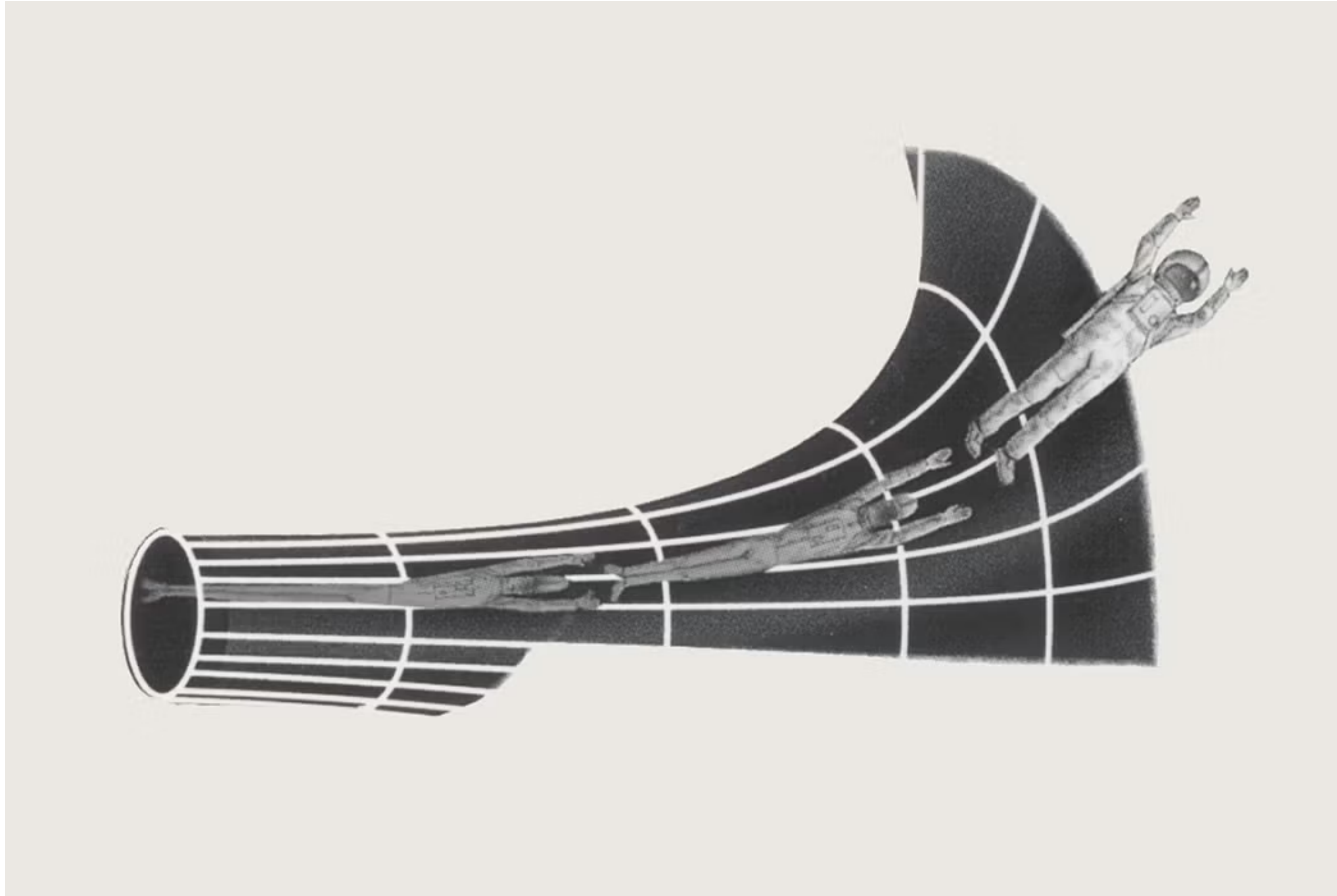
Does it make sense that the astronauts survived falling into Gargantua?



Spaghettification occurs when tidal forces lead to much stronger gravitational pull at the bottom of your feet than the top of your head

$$a_{\text{spaghetti}} = \frac{2GmL}{R^3}$$

Does it make sense that the astronauts
survived falling into Gargantua?



$$a_{\text{spaghetti}} = \frac{2GmL}{R^3}$$

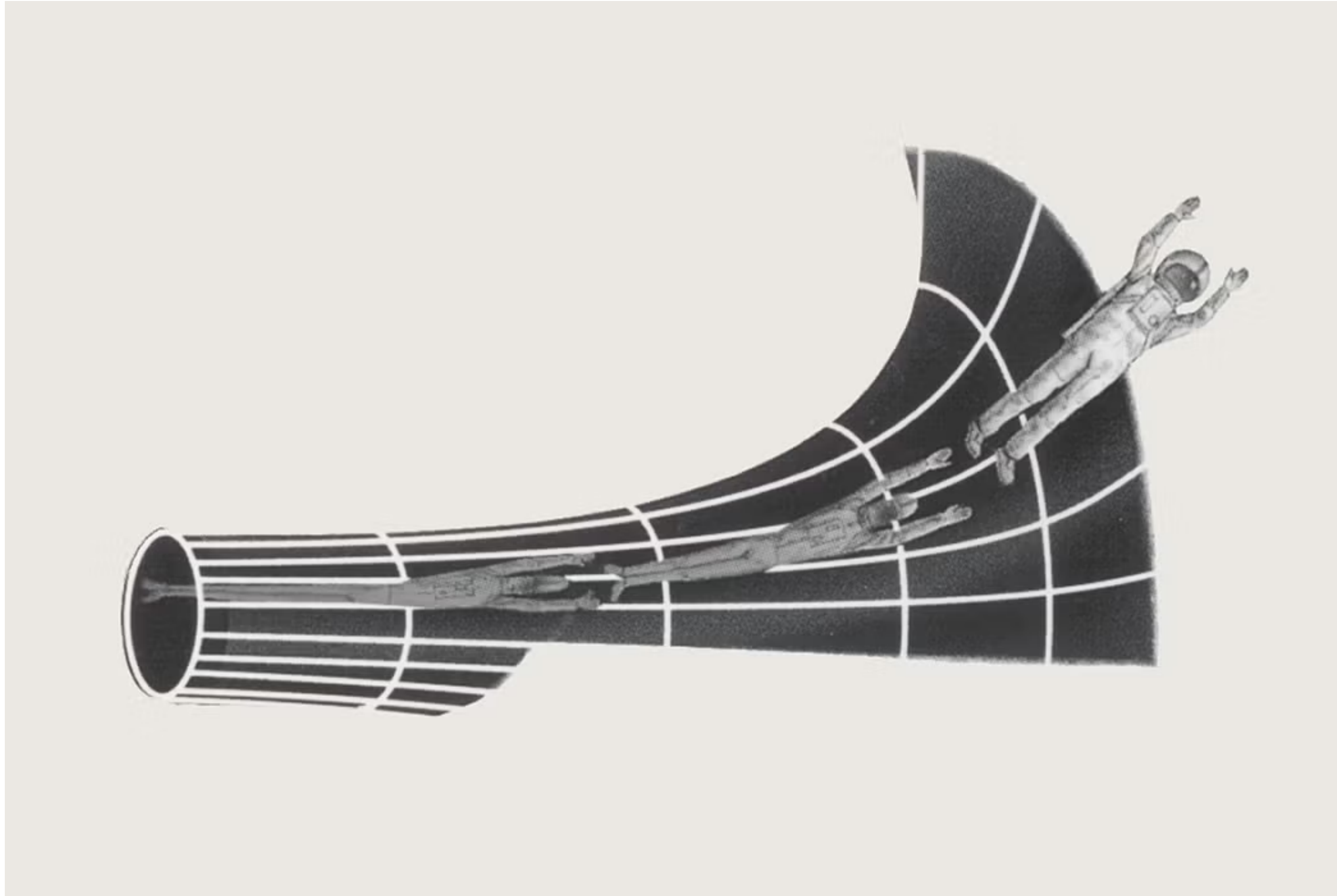
$$m = m_{\text{Gargantua}} = 10^9 M_{\odot}$$

$$L = 2m$$

$$a_{\text{spaghetti}} = 25 \text{ m/s}^2$$

$$R_{\text{spaghetti}} = \left(\frac{2GmL}{a_{\text{spaghetti}}} \right)^{1/3}$$

Does it make sense that the astronauts survived falling into Gargantua?



$$m = m_{\text{Gargantua}} = 10^9 M_{\odot}$$

$$L = 2\text{m}$$

$$a_{\text{spaghetti}} = 25 \text{ m/s}^2$$

$$R_{\text{spaghetti}} = \left(\frac{2GmL}{a_{\text{spaghetti}}} \right)^{1/3}$$

$$R_{\text{spaghetti}} = 2.8 \times 10^8 \text{ m}$$

Does it make sense that the astronauts
survived falling into Gargantua?

$$r_s = \frac{2Gm}{c^2} \qquad m_{\text{Gargantua}} = 10^9 M_{\odot}$$

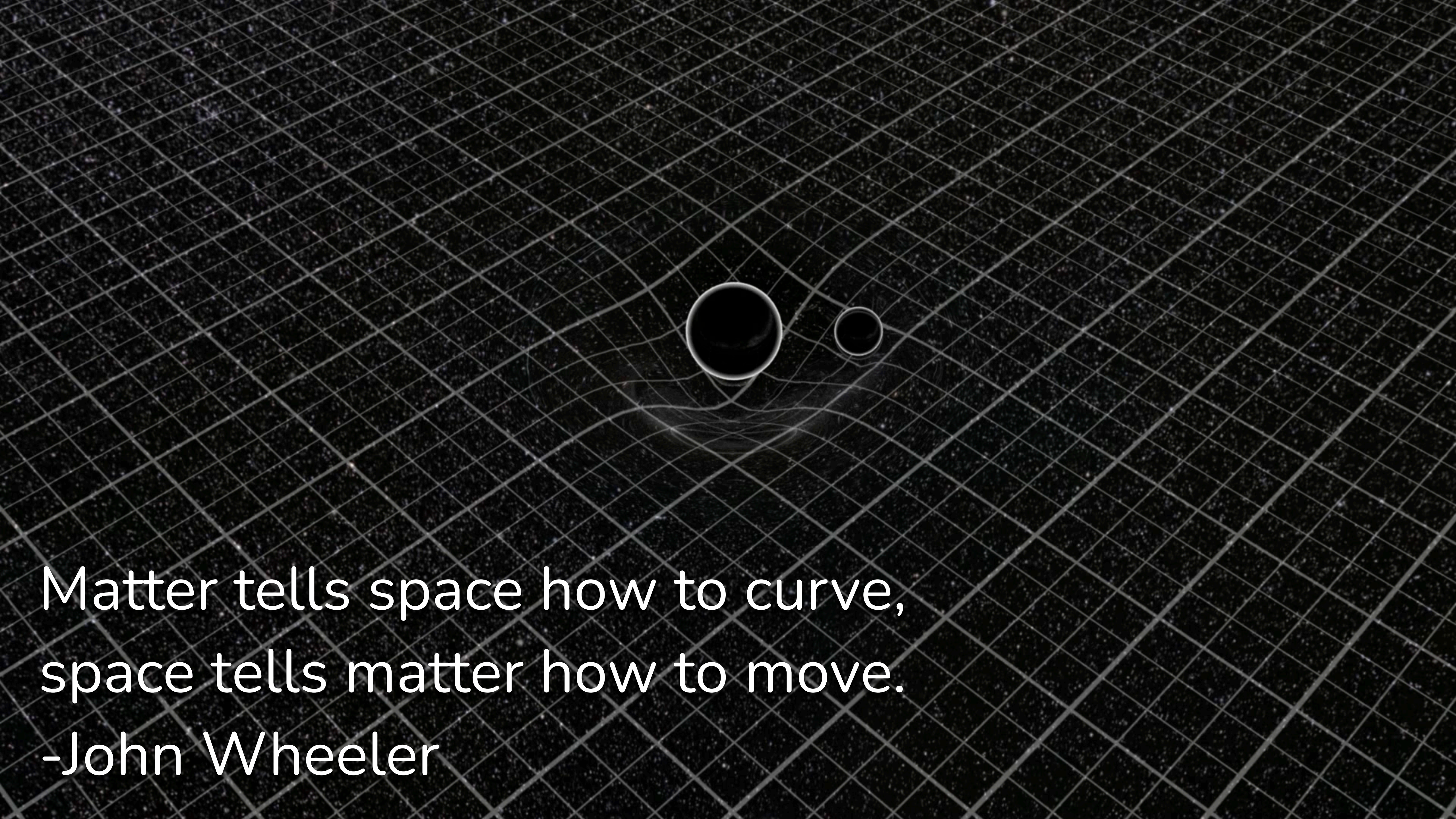
$$r_{s, \text{Gargantua}} = \frac{2Gm_{\text{Gargantua}}}{c^2} = \frac{2 \times 6.67 \times 10^{-11} \text{m}^3 \text{kg}^{-1} \text{s}^{-2} \times 10^9 \times 2 \times 10^{30} \text{kg}}{(3 \times 10^8 \text{m/s})^2}$$

$$r_{s, \text{Gargantua}} = 3 \times 10^{12} \text{m}$$

A million times larger than
the radius of the earth!

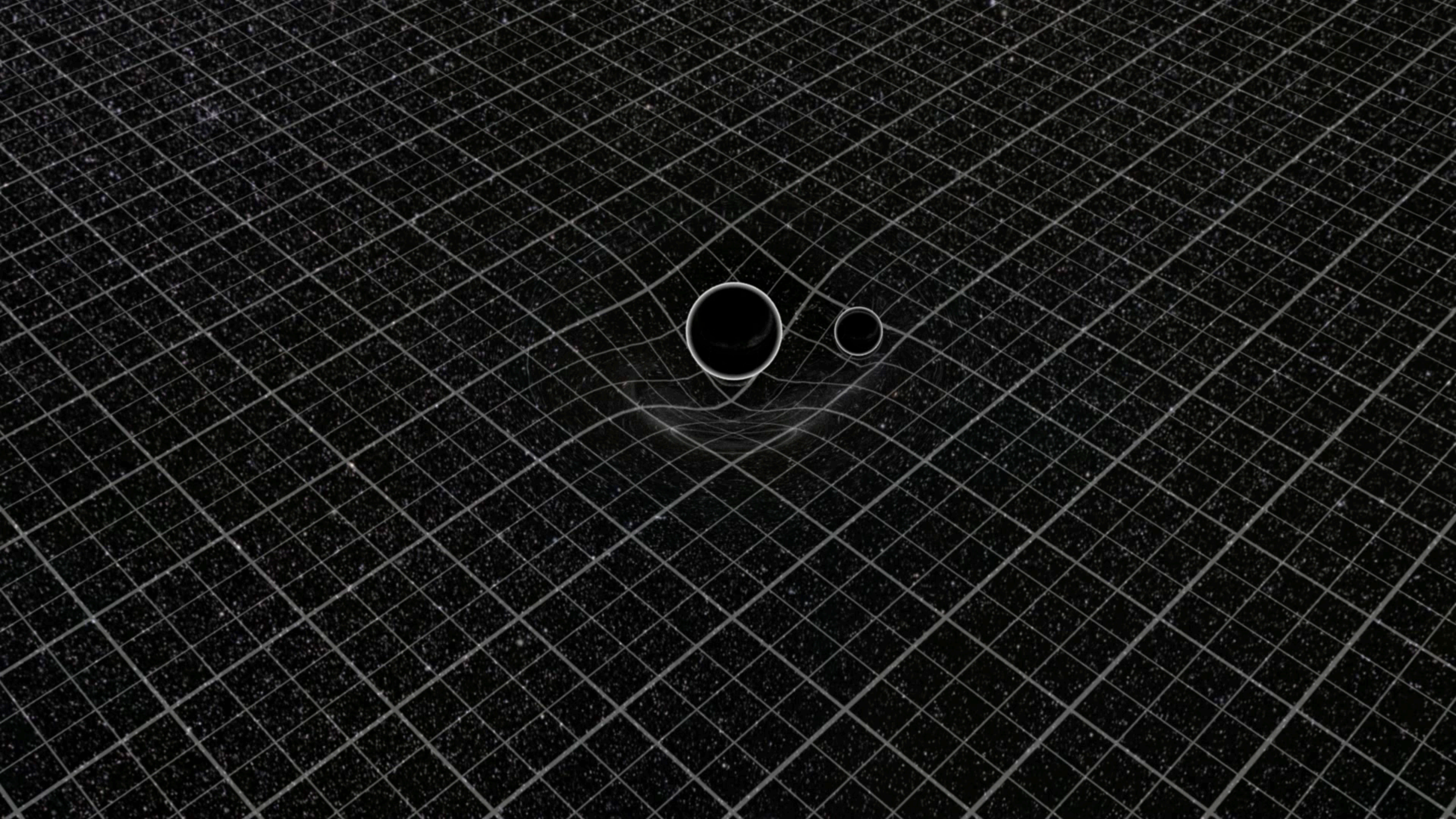
$$R_{\text{spaghetti}} = 2.8 \times 10^8 \text{m}$$

10,000 times larger than spaghettification!



Matter tells space how to curve,
space tells matter how to move.

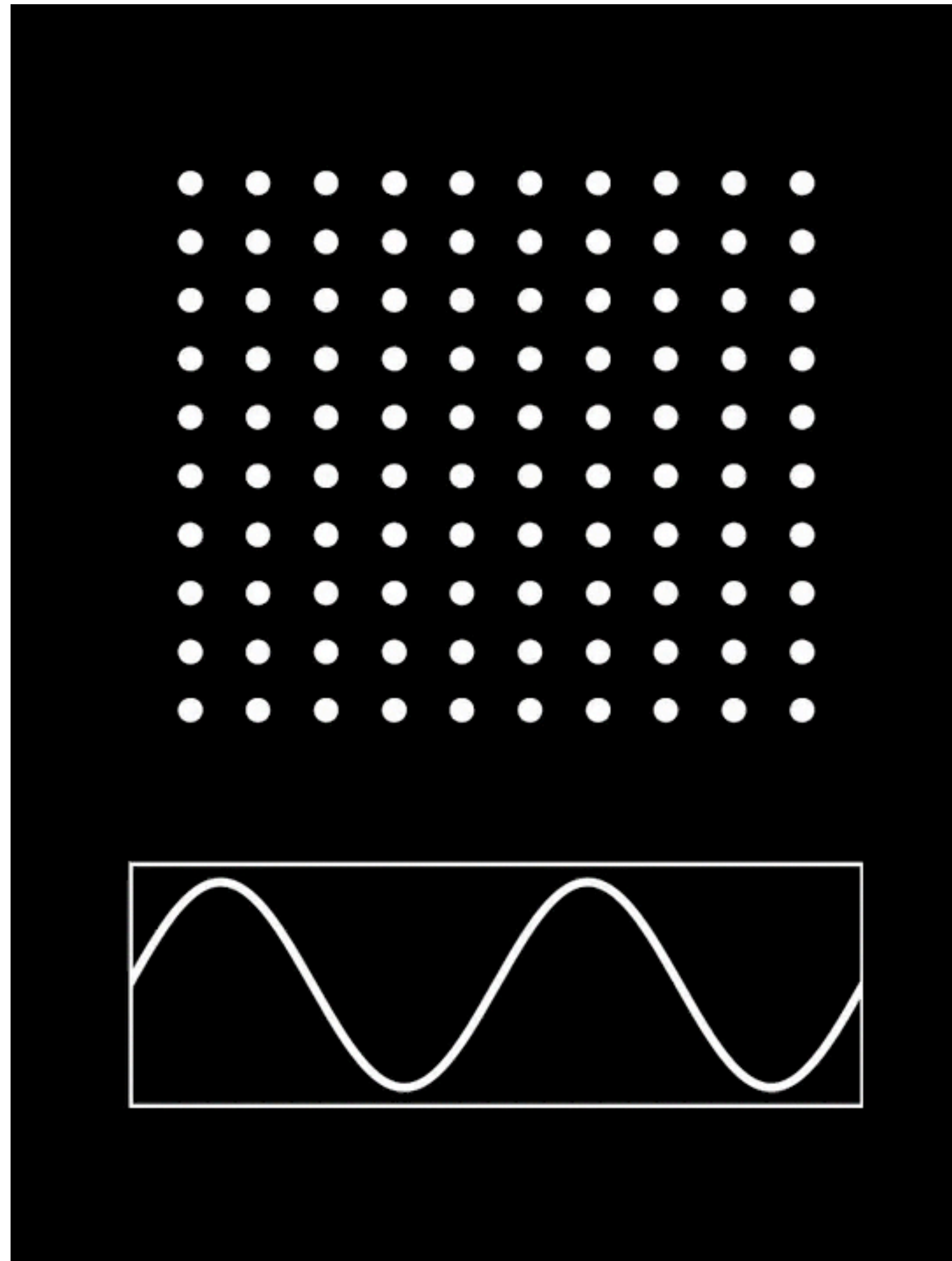
-John Wheeler





Star Trek VI: The Undiscovered Country

Directed:
Nicholas Meyer
Paramount (1991)



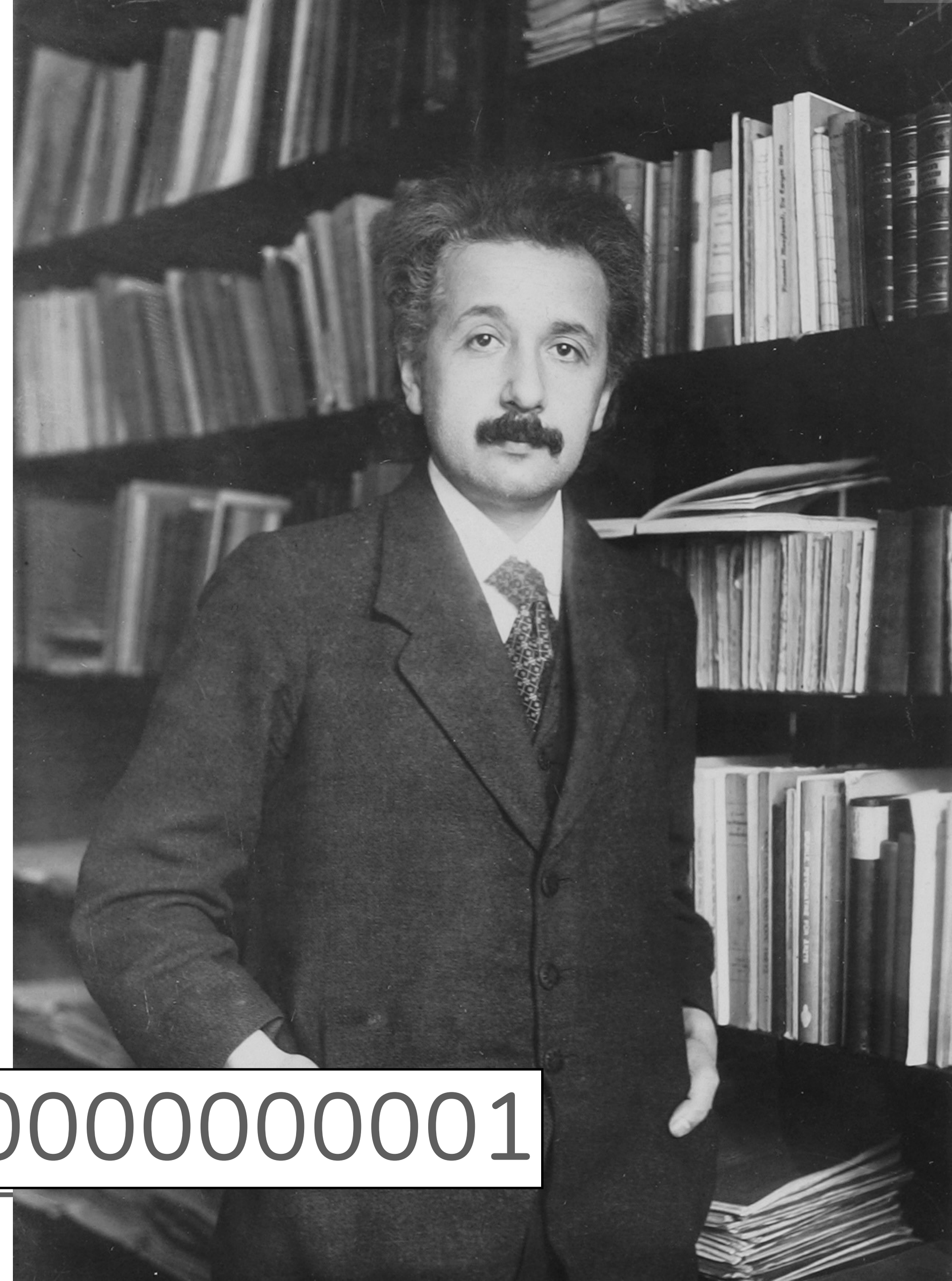
The ripples
stretch and
squash space
and time in
predictable ways

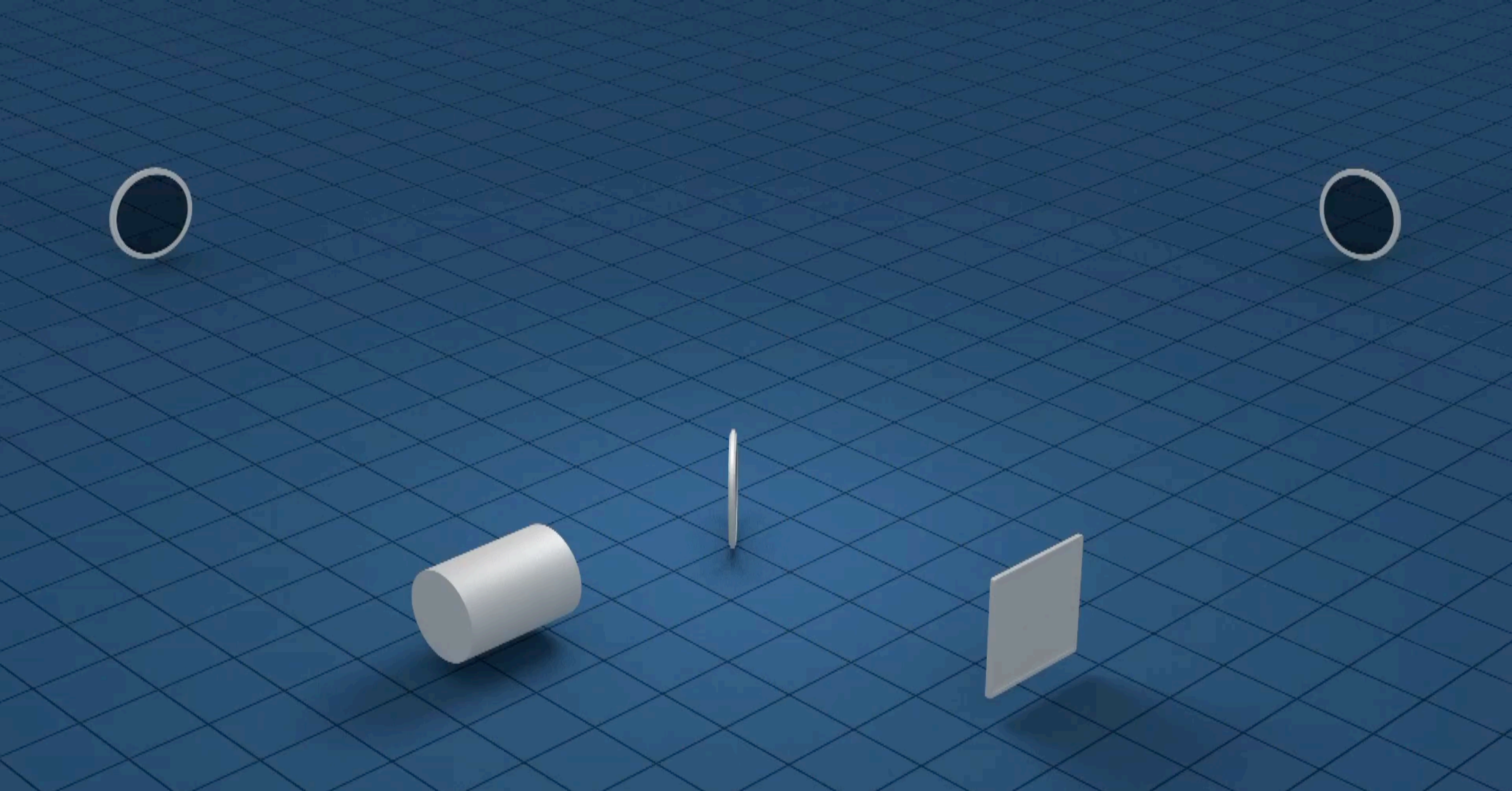
Einstein in 1916:

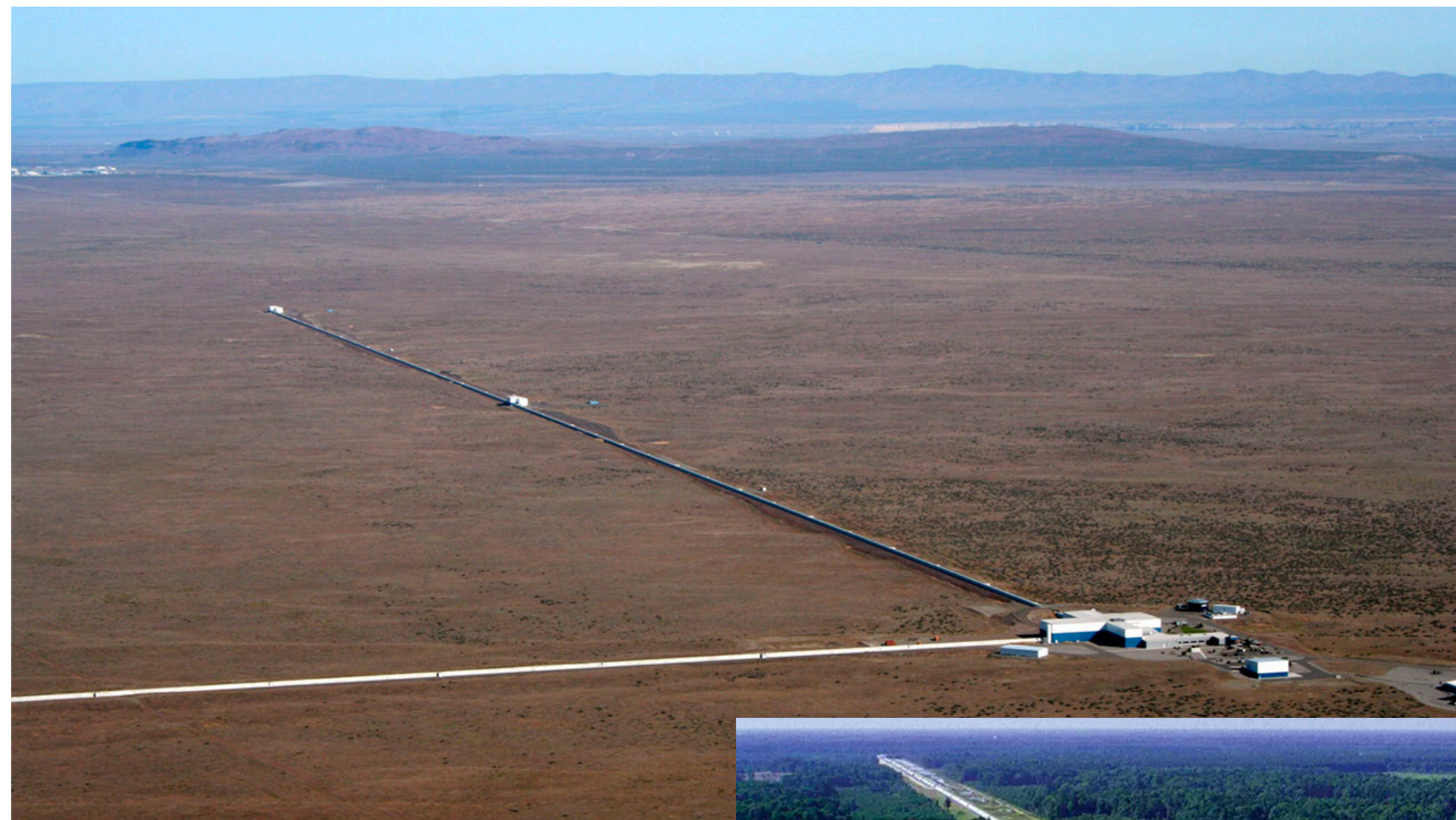
Gravitational waves exist
but they are way too
small to detect!

Size of stretches

$= 10^{-21}$

[illegible]





Washington

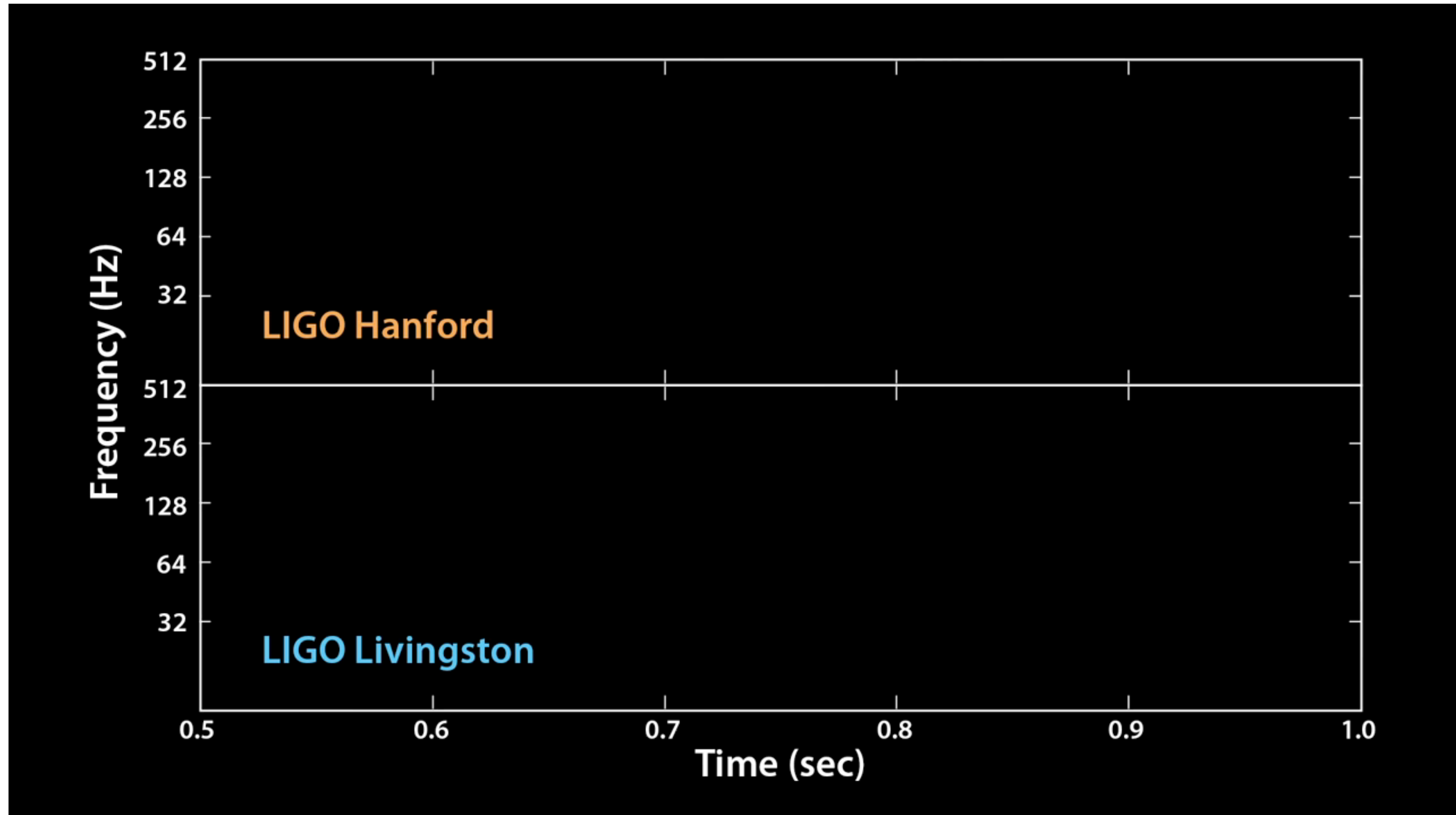


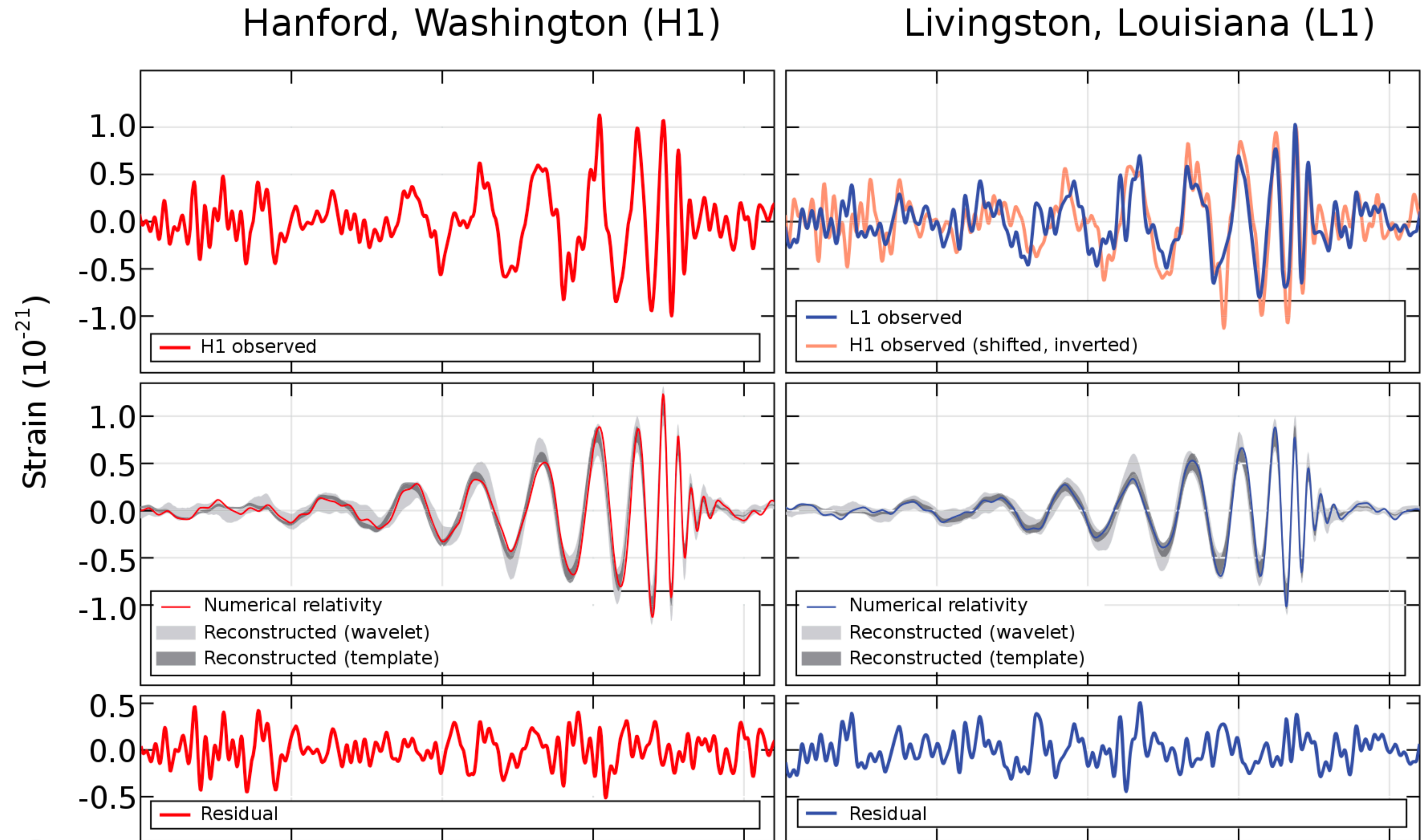
Louisiana



Italy

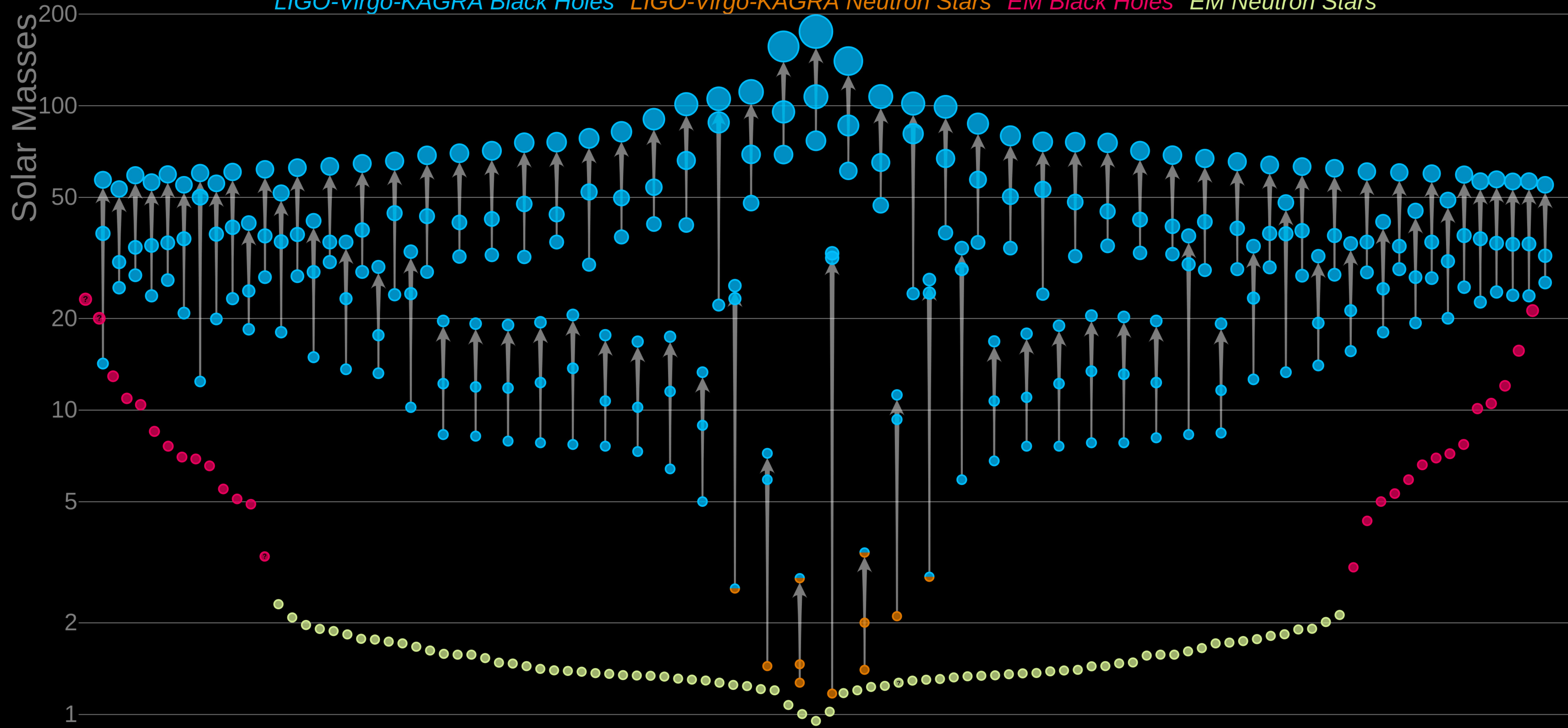
GW150914





Masses in the Stellar Graveyard

LIGO-Virgo-KAGRA Black Holes *LIGO-Virgo-KAGRA Neutron Stars* *EM Black Holes* *EM Neutron Stars*



 GraceDBPublic Alerts ▾LatestSearchDocumentationLogin

Please log in to view full database contents.

LIGO/Virgo/KAGRA Public Alerts

- More details about public alerts are provided in the [LIGO/Virgo/KAGRA Alerts User Guide](#).
- Retractions are marked in **red**. Retraction means that the candidate was manually vetted and is no longer considered a candidate of interest.
- Less-significant events are marked in **grey**, and are not manually vetted. Consult the [LVK Alerts User Guide](#) for more information on significance in O4.
- Less-significant events are not shown by default. Press "**Show All Public Events**" to show significant and less-significant events.

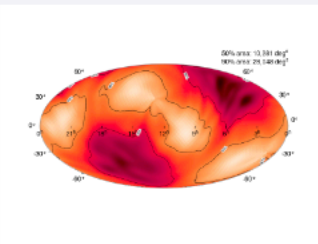
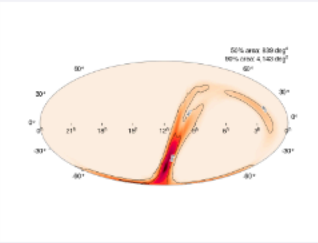
O4 Significant Detection Candidates: **81** (92 Total - 11 Retracted)

O4 Low Significance Detection Candidates: **1610** (Total)

Show All Public Events

Page 1 of 7. [next](#) [last](#) »

SORT: EVENT ID (A-Z) ▾

Event ID	Possible Source (Probability)	Significant	UTC	GCN	Location	FAR	Comments
S240109a	BBH (99%)	Yes	Jan. 9, 2024 05:04:31 UTC	GCN Circular Query Notices VOE		1 per 4.3136 years	
S240107b	BBH (97%), Terrestrial (3%)	Yes	Jan. 7, 2024 01:32:15 UTC	GCN Circular Query Notices VOE		1.8411 per year	

187 new events
in the fourth
observing run

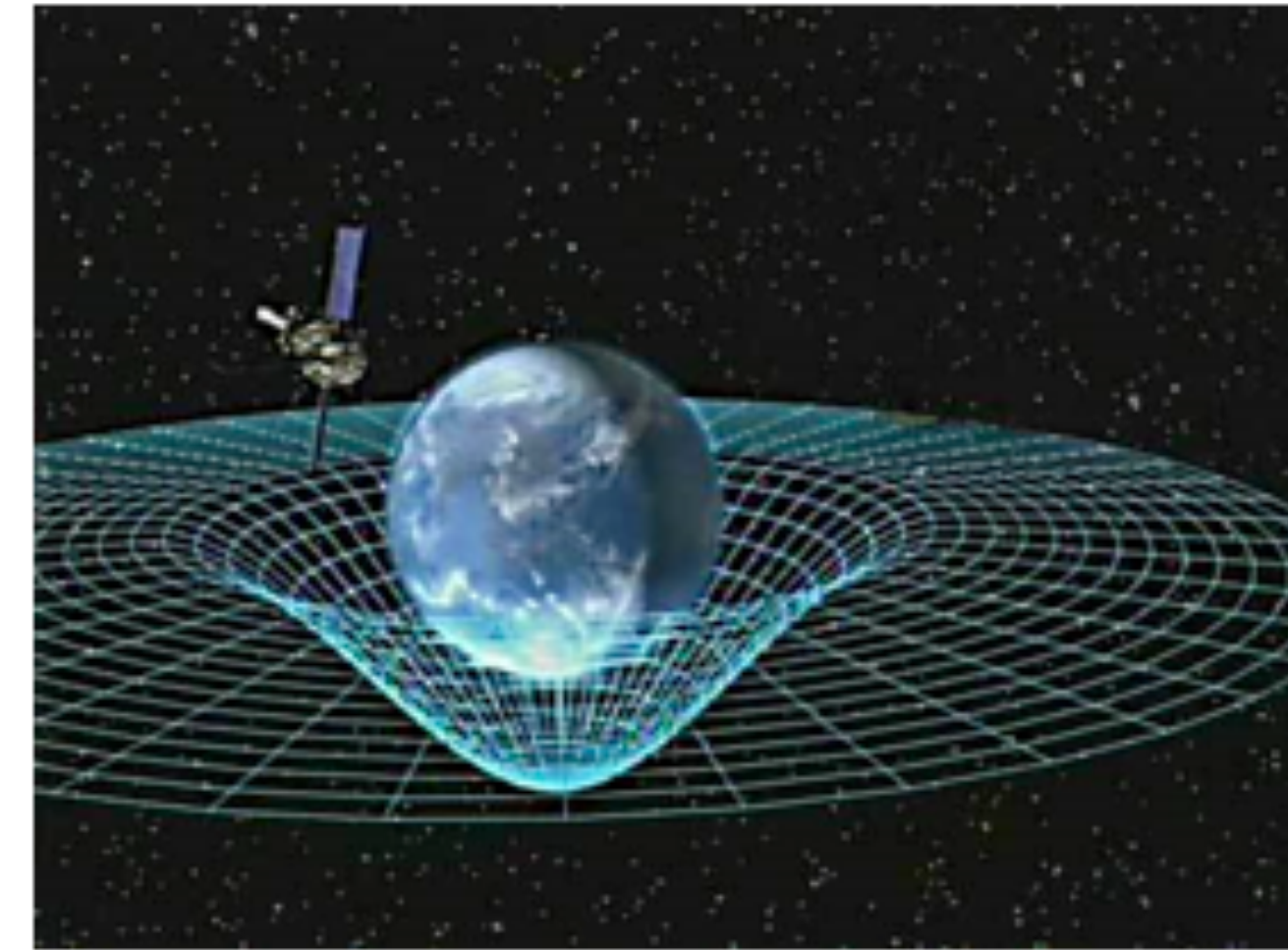
—
Nearing 250
events total!

The detectors
are currently on
until April 1 and
will be back on
June 7 - Oct 4

<https://gracedb.ligo.org/superevents/public/O4/>

Next time:

- ▶ Gravitational waves as a means of creating a warp drive
- ▶ Begin “future physics”



What is the nature of space and time?